

Assembly of a Cantenna-Based Radio Telescope and Observations of the 21cm Neutral Hydrogen Line

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ABSTRACT

While radio astronomy provides critical insights into cosmic structures, the high cost of conventional instrumentation remains a significant barrier for baseline research and educational platforms. Leveraging recent advancements in affordable Software Defined Radio (SDR) technology, this project demonstrates the development, calibration, and quantitative validation of a low-cost, cantenna-based radio telescope optimized for detecting the 21 cm (1.42 GHz) neutral hydrogen (H I) spectral line.

The experimental setup integrates a custom-built aluminum waveguide and an optimized horn extension with a high-gain, low-noise amplification chain. To isolate instrumental noise, the frequency-dependent system noise temperature (T_{sys}) was empirically calibrated via the Y-factor method, utilizing the blank background sky as a cold reference and a dense forest canopy approximated as a thermal blackbody. This calibration yielded a stable baseline with an operational mean system temperature of $\langle T_{\text{sys}} \rangle \approx 204$ K. Subsequent drift-scan observations were executed across a trajectory of decreasing Galactic latitude toward the Galactic equator.

The baseline-subtracted spectra revealed a systematic, non-linear rise in intensity, capturing a peak antenna temperature of $T_{\text{A, peak}} = 2.2$ K ($T_{\text{B, peak}} = 3.4$ K) at $b = 56.55^\circ$ that sharply escalated to a maximum of $T_{\text{A, peak}} = 24.0$ K ($T_{\text{B, peak}} = 37.4$ K) directly on the Galactic mid-plane ($b = 0.21^\circ$). Direct quantitative comparison with the standardized Leiden/Argentine/Bonn (LAB) Survey profiles confirmed excellent agreement. Ultimately, despite intrinsic constraints such as terrestrial sidelobe pickup, this study validates the scientific fidelity of ultra-low-cost radio frameworks, offering a highly robust, reproducible blueprint for hands-on radio astronomy and exploratory engineering.

1. INTRODUCTION

Radio astronomy is an important branch of observational astronomy that investigates celestial objects through their radio-frequency emissions. Unlike optical observations, radio observations can reveal physical processes and cosmic structures that are hidden at visible wavelengths. Since the discovery of cosmic radio emissions in the early twentieth century, radio astronomy has contributed significantly to our understanding of the Milky Way, distant galaxies, pulsars, and the large-scale structure of the universe.

Among the many radio signals observed in astronomy, the 21 cm neutral hydrogen line is one of the most critical. This spectral line originates from the hyperfine spin-flip transition of neutral hydrogen atoms and has a rest frequency of 1420.40575 MHz [Hellwig et al. \(1970\)](#). Because neutral hydrogen is the most abundant element and is widely distributed throughout the interstellar medium, observations of the 21 cm line provide invaluable information about the spiral structure and dynamics of the Milky Way. Consequently, this hydro-

gen line serves as a primary, accessible target for introductory and educational radio astronomy.

Professional radio telescopes are powerful scientific instruments, but their construction and operation require substantial financial and technical resources, posing a significant barrier for students and amateur researchers. In recent years, however, rapid advancements in semiconductor technology have led to the emergence of highly affordable Software Defined Radio (SDR) receivers. This technological shift has democratized the field, enabling low-cost, cost-effective radio astronomy observations using accessible commercial equipment.

In this study, a low-cost, cantenna-based radio telescope was designed and assembled utilizing commercially available RF components and an SDR receiver. The entire engineering process, including the specific cantenna construction and receiver configuration, is described in detail to serve as a practical, step-by-step manual for future educational initiatives. Using this assembled system, we successfully observed the 21 cm neutral hydrogen line. Through this project, we demonstrate that meaningful radio astronomy observations can be achieved with a minimal budget, providing an excel-

lent educational framework for learning the fundamental concepts and practical skills of observational radio astronomy.

2. CANTENNA-BASED RADIO TELESCOPE DESIGN AND THEORY

A cantenna can be conceptualized as a combination of a circular waveguide and a monopole probe antenna. Therefore, to optimize its performance for radio astronomy observations, the electromagnetic characteristics of both elements must be considered simultaneously.

Inside a circular waveguide, electromagnetic waves propagate in various modes, each characterized by a specific cutoff frequency (f_c). If multiple modes propagate concurrently, mode coupling and interference occur, leading to significant signal loss and distortion. To prevent this, the waveguide must be designed to support only a single, dominant mode at the target frequency of 1.42 GHz.

According to Pozar [Pozar \(2012\)](#), the cutoff frequency for a specific mode in a circular waveguide of inner diameter D is given by:

$$f_c = \frac{c \cdot x}{\pi D} \quad (1)$$

where c is the speed of light (3.00×10^8 m/s) and x is the eigenvalue (the roots of the derivative of the Bessel function) for the respective mode.

Equation (1) demonstrates that the cutoff frequency is a direct function of the waveguide diameter. In a circular waveguide, the dominant mode—which has the lowest cutoff frequency—is the TE_{11} mode with an eigenvalue of $x = 1.841$ [Pozar \(2012\)](#). The next higher-order mode is the TM_{01} mode, where $x = 2.405$. To ensure single-mode propagation, the operational frequency must lie strictly between the cutoff frequencies of these two modes. By rearranging Equation (1) in terms of the diameter D , we obtain the physical dimension constraint:

$$D = \frac{c \cdot x}{\pi \cdot f_c} \quad (2)$$

To determine the allowable physical dimensions for observing the 21 cm neutral hydrogen line ($f = 1.42$ GHz), the minimum diameter ($D_{\min}$) required to allow the TE_{11} mode to propagate and the maximum diameter ($D_{\max}$) allowed before the secondary TM_{01} mode begins to propagate are calculated as:

$$D_{\min} = \frac{3.00 \times 10^8 \times 1.841}{3.14 \times 1.42 \times 10^9} \approx 124 \text{ mm} \quad (3)$$

$$D_{\max} = \frac{3.00 \times 10^8 \times 2.405}{3.14 \times 1.42 \times 10^9} \approx 162 \text{ mm} \quad (4)$$

Consequently, the physical condition for maintaining stable single-mode propagation within the cantenna at 1.42 GHz is established as:

$$124 \text{ mm} < D < 162 \text{ mm} \quad (5)$$

Any commercially acquired metal can utilized for this project must satisfy this diameter range to ensure an optimal signal-to-noise ratio during observations.

Inside the circular waveguide, electromagnetic waves propagate by reflecting off the inner conducting walls, which causes the guided wavelength (λ_g) to become longer than the free-space wavelength (λ_0). According to Pozar [Pozar \(2012\)](#), this relationship is defined as:

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{f_c}{f_0}\right)^2}} = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}} \quad (6)$$

where f_0 is the operational frequency (1.42 GHz) and f_c is the cutoff frequency of the dominant TE_{11} mode.

To maximize the signal-to-noise ratio of the weak 21 cm neutral hydrogen line, the physical dimensions of the cantenna must be optimized based on these wavelengths. The schematic diagram of the designed cantenna is illustrated in Figure 1.

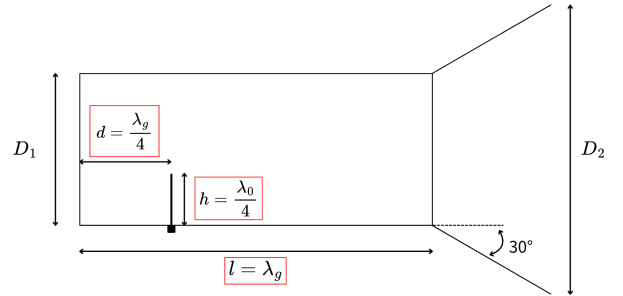


Figure 1. Side view schematic and geometric parameters of the combined cantenna and horn antenna system.

First, the optimal probe position (d), which represents the distance from the closed backplate of the can to the monopole probe, is determined by:

$$d = \frac{\lambda_g}{4} \quad (7)$$

This quarter-guided-wavelength spacing ensures that the waves reflected from the short-circuited backplate undergo a 180° phase shift and constructively interfere with the incident waves, establishing a standing wave peak precisely at the probe's location.

Second, the optimal probe height (h) inside the waveguide is designated as:

$$h = \frac{\lambda_0}{4} \quad (8)$$

Since the probe functions essentially as a quarter-wave monopole antenna to couple the transverse electric fields, its physical height must be sized relative to the free-space wavelength (λ_0) rather than the guided wavelength (λ_g).

Finally, the total length of the waveguide (l) must satisfy the following minimum condition:

$$l \geq \lambda_g \quad (9)$$

A total length of at least one guided wavelength is critically required to allow the internal standing waves to form stably without edge-effect distortions at the open aperture.

To further enhance the directivity and gain of the antenna for detecting the weak cosmic signal, an optimum pyramidal or conical horn extension was designed. According to Balanis [Balanis \(2005\)](#), the optimum aperture diameter (D_2) of a horn antenna is given by:

$$D_2 = \sqrt{3 \cdot \lambda_0 \cdot (D_2 - D_1)} \quad (10)$$

where λ_0 is the free-space wavelength of the 21 cm line and D_1 is the inner diameter of the existing antenna waveguide.

The layout of the designed horn are illustrated in Figure 2.

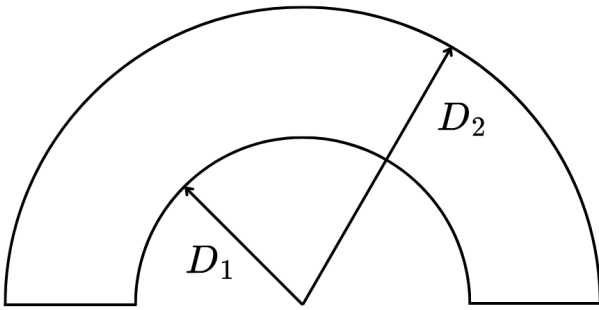


Figure 2. Flat pattern development drawing for the fabrication of the optimum horn antenna.

To solve for the optimum aperture diameter D_2 , Equation (10) can be rearranged into a standard quadratic form as follows [Balanis \(2005\)](#):

$$D_2^2 - 3\lambda_0 D_2 + 3\lambda_0 D_1 = 0 \quad (11)$$

By substituting the measured antenna diameter D_1 and the free-space wavelength λ_0 at 1.42 GHz into Equation (11), the smaller real root was selected to determine the physically optimized and compact aperture dimensions for the horn structure.

3. PREPARATION FOR ANTENNA-BASED RADIO TELESCOPE ASSEMBLY

Before commencing the physical assembly of the radio telescope, all necessary components, radio frequency (RF) electronics, and structural assembly tools were prepared based on the theoretical design requirements. The component specifications and system configuration designed for the radio telescope are detailed below:

3.1. RF Electronics

- **Aluminum Can (Qty: 1):** Functions as the main waveguide body with an inner diameter of approximately 152 mm.
- **Nooelec NESDR SMARt v5 (Qty: 1):** Main SDR receiver supporting a frequency range of 100 kHz to 1.75 GHz (NF $\approx$ 10 dB).
- **Nooelec SAWbird+ H1 (Qty: 1):** Pre-stage Low Noise Amplifier providing $\sim$ 40 dB gain (NF $\approx$ 0.8 dB) with an integrated 1.42 GHz band-pass filter.
- **SPF5189Z (Qty: 1):** Post-stage Low Noise Amplifier providing an additional $\sim$ 20 dB gain (NF $\approx$ 0.8 dB).
- **Copper Tube & SMA Probe Connector (Qty: 1 each):** Monopole antenna element (inner diameter $\approx$ 2 mm) and connector for RF coupling.
- **SMA Coaxial Cable (Qty: 3–4):** Low-loss RF interconnects adjusted to structural requirements.

3.2. Assembly Tools

- **Structural Materials:** Heavy-duty aluminum grill foil for horn lining, DIY solderable DC power cable, USB-to-DC voltage supply cable, and M2.5 bolt/nut sets (2 sets).
- **Hardware Tools:** A comprehensive kit including a drill, hacksaw, cross screwdriver, ruler, utility knife, scissors, tape, name pen, and a complete soldering setup.

4. STEP-BY-STEP ASSEMBLY MANUAL

This section describes the practical assembly sequence of the antenna-based radio telescope. All physical di-

mensions are grounded in the theoretical relations derived in Section 2 to ensure structural and operational precision.

4.1. Waveguide Body Drilling

The optimal probe position from the short-circuited backplate was measured and marked at $d = 90.8$ mm according to Equation (7). A feed hole was then drilled into the aluminum can using a step drill bit. To ensure precise alignment without offset errors, it is recommended to pre-mark the mounting screw holes by placing the SMA connector flange directly onto the can before drilling. Care must be taken to control the drill speed, as excessive force or velocity can easily deform or tear the thin aluminum wall.



Figure 3. Drilling process and completed feed hole on the waveguide body.

4.2. Monopole Probe Cutting

A solid copper tube was cut using a hacksaw to the theoretical quarter-wavelength height of $h = 52.8$ mm derived from Equation (8). Scoring a light, consistent baseline around the tube circumference prior to cutting serves as an effective guide for a clean saw line. Protective gloves must be worn during this process to prevent lacerations from sharp metal edges and latent burrs. For finer impedance tuning and matching, this length can be slightly reduced to 46–47 mm depending on the internal boundaries.

4.3. Probe Soldering

The cut copper tube was joined to the center pin of the SMA connector and permanently secured via soldering. Standard thermal precautions must be exercised to prevent skin burns from the iron.

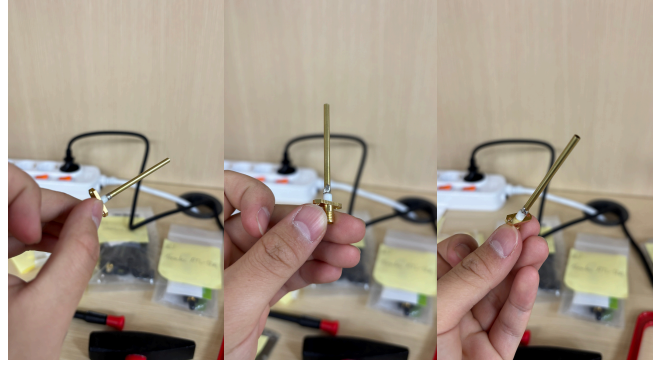


Figure 4. Soldered SMA monopole probe assembly.

4.4. Probe and Waveguide Integration

The completed SMA probe assembly was inserted from the inside of the aluminum can and structurally fastened through the drilled feed hole using M2.5 bolts and nuts.

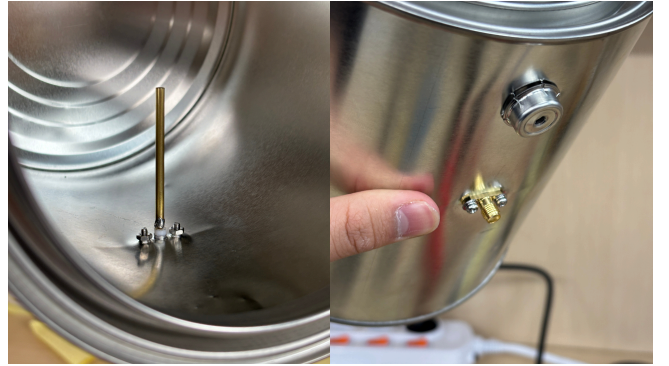


Figure 5. Internal and external views of the integrated probe assembly.

4.5. LNA Power Harnessing

To supply power to the SPF5189Z LNA, a DIY DC power cable was soldered directly to the board terminals. For proper polarity matching, the positive (+) wire was secured to the VCC terminal, and the negative (-) wire was secured to the GND terminal. Applying a small amount of soldering flux to the stripped strands significantly improves solder wetting, though thermal exposure to the PCB pads should be minimized to prevent trace delamination.

4.6. Waveguide Extension Barrel Fabrication

An extension barrel was constructed using cardboard boxing lined with heavy-duty aluminum foil to meet the single-mode waveguide length requirement of $l = 363$ mm from Equation (9).

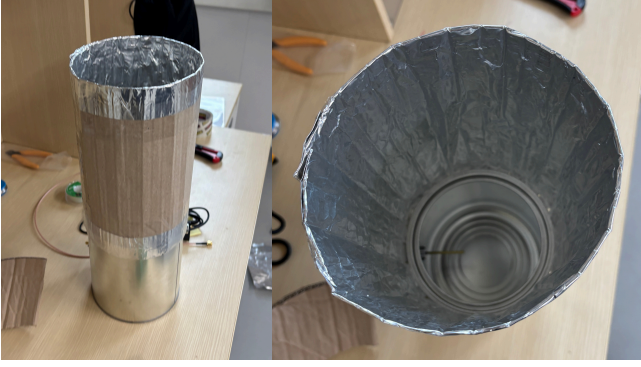


Figure 6. Assembled aluminum-lined extension barrel.

4.7. Optimum Horn Antenna Construction

The flat pattern pattern layout of the horn extension was sketched onto cardboard based on the aperture dimensions from Equations (10) and (11), then cut out and lined with aluminum foil. The final horn assembly was securely mated to the extension barrel using tape.

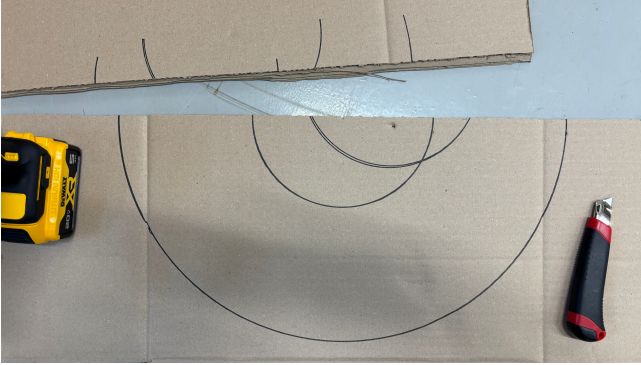


Figure 7. Geometrical sketching of the horn pattern.



Figure 8. Side and top views of the completed horn antenna.

4.8. Step 8: Final RF Receiver Chain Integration

The active RF frontend components were linked sequentially using low-loss SMA coaxial cables. The signal path flows from the cantenna probe $\rightarrow$ Nooelec SAWbird+ H1 $\rightarrow$ SPF5189Z LNA $\rightarrow$ Nooelec NESDR SMART v5 receiver. Because RF amplifiers are highly directional, the Input (IN) and Output (OUT) ports of each module must be strictly verified before final connection to prevent signal blockage or component damage.

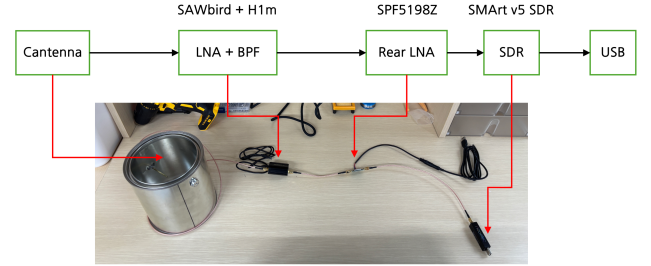


Figure 9. Complete RF receiver chain layout.

5. OBSERVATION METHODOLOGY

This section outlines the theoretical framework and signal processing pipeline established to calibrate the raw power measurements and derive astronomical physical quantities. The data reduction sequence flows from receiver temperature to the final calculation of brightness temperature via the Rayleigh–Jeans approximation.

5.1. System Noise Calibration

To isolate the cosmic signal from instrumental effects, the noise temperature of the receiver (T_{rec}) must be determined via the classical Y-factor method. The primary power dissipation follows the Johnson–Nyquist noise relation:

$$P = Gk_{\text{B}}T\Delta f \quad (12)$$

where G is the overall system gain, k_{B} is the Boltzmann constant, and Δf is the operational bandwidth. By measuring the power ratio between a known hot load (P_{hot}) and a cold load (P_{cold}), the Y-factor is established as:

$$Y = \frac{P_{\text{hot}}}{P_{\text{cold}}} = \frac{T_{\text{rec}} + T_{\text{hot}}}{T_{\text{rec}} + T_{\text{cold}}} \quad (13)$$

Rearranging this relation yields the receiver noise temperature (Burke et al. 2019):

$$T_{\text{rec}} = \frac{T_{\text{hot}} - YT_{\text{cold}}}{Y - 1} \quad (14)$$

In this setup, a dense forest canopy was utilized as the hot load ($T_{\text{hot}} \approx T_{\text{ambient}}$). Due to the complex,

multi-layered structure of the foliage, the canopy acts as an effective radio absorber with negligible reflectivity, providing a sound physical justification for approximating it as an ambient blackbody source. Conversely, the blank background sky served as the cold load. Following standard empirical atmospheric models, the baseline cold temperature was defined as $T_{\text{cold}} = T_{\text{CMB}} + T_{\text{Galaxy}} + T_{\text{atmosphere}} = 7 \text{ K}$ (Burke et al. 2019).

The total system noise temperature (T_{sys}) encompasses both internal instrumental noise and external environmental contributions, expressed as:

$$T_{\text{sys}} = T_{\text{CMB}} + T_{\text{Galaxy}} + T_{\text{atmosphere}} + T_{\text{sidelobe}} + T_{\text{rec}} \quad (15)$$

Accounting for additional sidelobe pick-up and ground noise contributions ($\sim 10 \text{ K}$), the total system noise temperature simplifies to:

$$T_{\text{sys}} = T_{\text{rec}} + 17 \text{ K} \quad (16)$$

5.2. Antenna Temperature and Flux Density

The raw observational data consists of targeted power spectra on the Galactic plane (P_{on}) and reference spectra centered on the blank sky (P_{off}). The uncalibrated antenna temperature (T_{A}) is extracted by scaling the relative power fractional change with the calculated system temperature (Issaoun et al. 2017):

$$T_{\text{A}} = \frac{P_{\text{on}} - P_{\text{off}}}{P_{\text{off}}} \cdot T_{\text{sys}} \quad (17)$$

To map the measured antenna temperature to a physical flux density (S) in Janskys (Jy), the effective aperture area (A_{eff}) of the horn antenna must be incorporated:

$$S = T_{\text{A}} \cdot \frac{2k_{\text{B}}}{A_{\text{eff}}} \quad (18)$$

The effective area scales with the physical geometric aperture area (A_{phy}) through the aperture efficiency (η) (Issaoun et al. 2017):

$$A_{\text{eff}} = \eta A_{\text{phy}} = \eta \pi \left(\frac{D}{2} \right)^2 \quad (19)$$

where D represents the physical diameter of the antenna horn aperture. For absolute SI compliance, the Boltzmann constant is strictly fixed to $k_{\text{B}} = 1.380649 \times 10^{-23} \text{ J/K}$ (Bureau International des Poids et Mesures (BIPM) 2019), or equivalently scaled to $1380.649 \text{ m}^2 \text{ Jy/K}$ to directly yield flux density units.

5.3. Brightness Temperature Estimation via Rayleigh-Jeans Approximation

The ultimate astronomical quantity representing the source intensity is the brightness temperature (T_{B}). In

the radio regime ($h\nu \ll k_{\text{B}}T$), the Planck radiation law reduces to the linear Rayleigh-Jeans approximation, allowing T_{B} to be expressed as a function of the flux density and the beam solid angle (Ω) (Condon & Ransom 2015):

$$T_{\text{B}} = \frac{S}{\Omega} \cdot \frac{\lambda^2}{2k_{\text{B}}} \quad (20)$$

Assuming a symmetric Gaussian beam profile, the beam solid angle (Ω) is approximated from the Half-Power Beam Width (θ_{HPBW}) as:

$$\Omega = \frac{\pi}{4 \ln(2)} \cdot \theta_{\text{HPBW}}^2 \approx 1.133 \theta_{\text{HPBW}}^2 \quad (21)$$

The antenna angular resolution, defined by the half-power beam width for a uniformly illuminated circular aperture, scales directly with the operational wavelength (λ) and the physical aperture diameter (D) (Condon & Ransom 2015):

$$\theta_{\text{HPBW}} \approx 1.2 \frac{\lambda}{D} \quad (22)$$

By executing this sequential calibration chain, raw power spectra digitized via the SDR receiver are successfully converted into calibrated physical brightness temperatures of neutral hydrogen lines.

5.4. Observational Environment

Observations were conducted at a parking lot on the UNIST campus (Latitude: 35.570387°N , Longitude: $129.185851^\circ \text{E}$). The site provided a clear, unobstructed view of the sky without any significant local obstacles. To characterize the surrounding environment, panoramic profiles of the terrain were recorded across the four cardinal sectors. Notably, a dense, homogeneous forest canopy in the southeastern sector was selected as the reference ambient blackbody source (T_{hot}) for the Y-factor calibration.

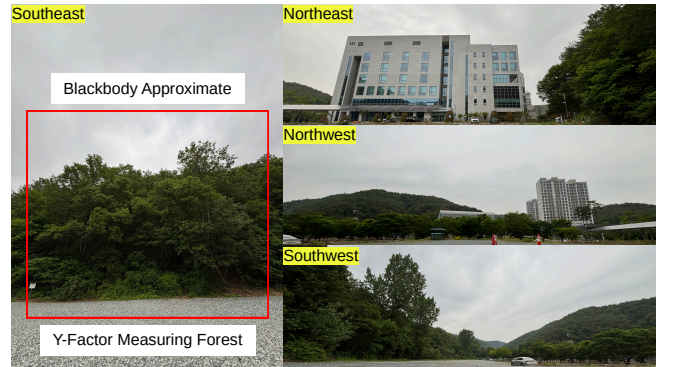


Figure 10. Panoramic surveyed profiles of the observation site, with the southeastern forest canopy highlighted as the reference hot-load blackbody.

The physical deployment of the telescope involved two distinct orientations depending on the operational phase. For primary astronomical data collection, the antenna was fixed in a zenith-pointing configuration to perform a drift-scan observation of the transiting sky. For calibration intervals, the antenna structure was manually slewed off-zenith to directly target the south-eastern forest canopy.



Figure 11. Operational configurations of the cantenna radio telescope showing the zenith-pointing drift-scan mode (left) and the off-zenith Y-factor calibration mode targeting the forest (right).

5.5. Data Acquisition Protocol and SDR Configuration

The targeted observation spanned a continuous 7-hour window from 22:00 KST on May 23, 2026, to 05:00 KST on May 24, 2026. The RF signal digitized via the Nooelec NESDR SMart v5 receiver was governed by a strict instrumentation protocol to maximize the signal-to-noise ratio (SNR) while avoiding non-linear artifacts.

The Software Defined Radio (SDR) gain was optimized and locked at 14.4 dB after verifying the complete absence of ADC saturation or clipping in the raw power domain. The center frequency was tuned to $\nu_0 = 1420.4$ MHz to encapsulate the neutral hydrogen (H I) rest frequency (1420.405 MHz) within a sampling rate of 2.048 MHz. Digital signal processing was executed with an FFT size of 1024 points, yielding a spectral frequency resolution of $\Delta\nu = 2$ kHz.

To smooth out stochastic thermal fluctuations, incoming spectra were integrated over 10-second blocks. The systematic observation timeline was scheduled as follows:

- **Zenith Science Scan:** Automated spectra acquisition was performed at regular 1-minute intervals centered at the zenith.
- **Calibration Interleaves:** Every 10 minutes, the telescope was temporarily slewed to sample the forest canopy for Y-factor cross-calibration. The

ambient temperature of the forest was recorded during each interleave to dynamically update T_{hot} in Equation (3).

Throughout the entire 7-hour campaign, the system was continuously monitored on-site to ensure structural integrity, prevent cable strain, and verify real-time data logging stability.

6. OBSERVATIONAL RESULTS

This section presents the empirical outcomes of the 7-hour observational campaign and evaluates the quantitative performance of the calibrated cantenna radio telescope. A total of 400 science data points targeting the celestial sphere and 40 calibration data points targeting the terrestrial forest canopy were successfully acquired. To ensure high spectral fidelity and remove transient Radio Frequency Interference (RFI) or instrumental anomalies, five prime subsets of baseline co-observed pairs (designated Obs 1 through Obs 5) were meticulously isolated for full calibration and system temperature derivation.

The five selected reference fields represent cold, blank sky regions chosen to accurately measure the baseline power (P_{off}) without local Galactic continuum contamination. The precise observation timestamps and corresponding coordinates mapped into the Galactic coordinate system are structured in Table 1.

To visualize the distribution of these calibration points relative to the main structure of our Galaxy, the spatial projection of the cold sky regions was tracked globally. As illustrated in Figure 12, the sequential pointing track of the drift-scan drifts away from the North Galactic Pole down toward the mid-latitude regimes, providing a robust, emission-free baseline profile.

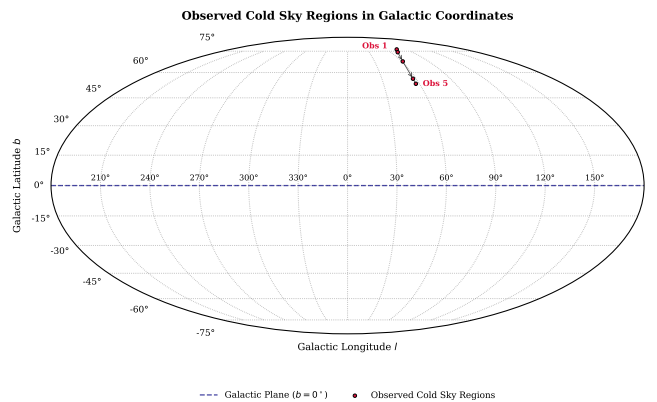


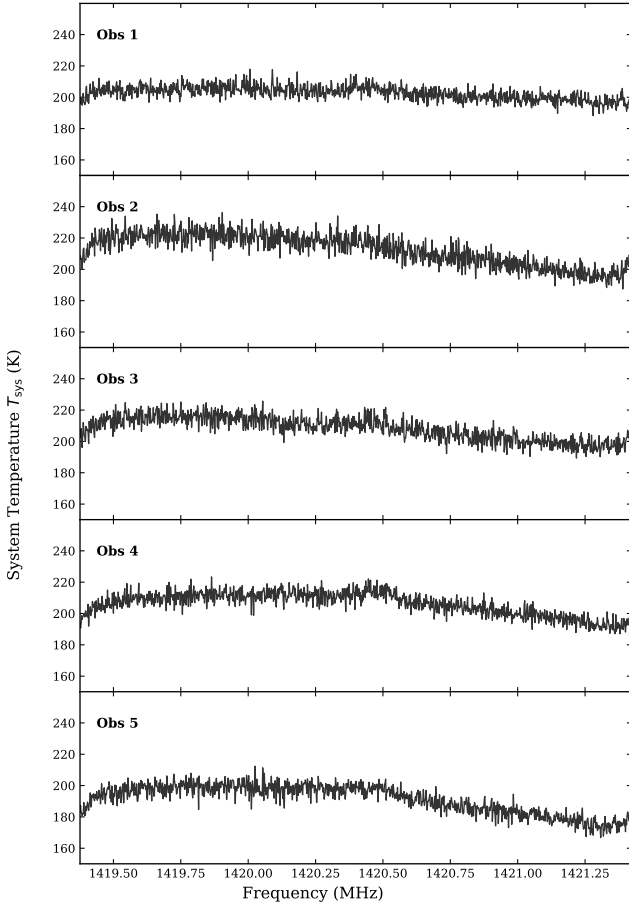
Figure 12. Spatial distribution of the five selected cold sky reference regions mapped in the Galactic coordinate system.

Table 1. Chronological Logging and Galactic Coordinates of Selected Reference Fields

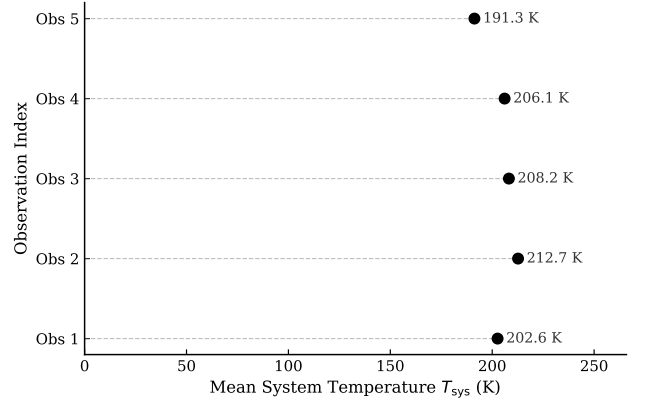
Index	Forest Obs (KST)	Blank Sky Obs (KST)	Galactic Longitude l ($^{\circ}$)	Galactic Latitude b ($^{\circ}$)
Obs 1	2026-05-23, 22:00:12	2026-05-23, 22:00:30	75.2523139200	76.4609620289
Obs 2	2026-05-23, 22:09:53	2026-05-23, 22:13:30	69.6568827859	74.2176495663
Obs 3	2026-05-23, 22:49:53	2026-05-23, 22:49:30	61.2796308249	67.3986363380
Obs 4	2026-05-23, 23:39:53	2026-05-23, 23:45:30	57.2728519965	56.1427384210
Obs 5	2026-05-23, 23:59:53	2026-05-24, 00:00:30	57.0408854589	53.0874208607

6.1. System Noise Temperature Characterization

By processing the uncorrupted P_{hot} and P_{off} pairings through the mathematical framework established in Section 4, the frequency-dependent system noise temperature (T_{sys}) was extracted for each individual run. Figure 13 displays the high-resolution spectral profiles across the full 2.048 MHz bandwidth for all five reference iterations. The individual profiles reveal highly consistent baseline patterns, demonstrating excellent gain stability over time despite minor high-frequency fluctuations.

**Figure 13.** High-resolution system temperature (T_{sys}) spectral profiles evaluated independently across the five selected observation sessions.

To assess the macro-scale variation of the system over the 7-hour span, the integrated mean system temperature for each block was calculated and compared. As summarized in Figure 14, the mean system temperature ranges from a minimum of 191.3 K (Obs 5) to a maximum of 212.7 K (Obs 2), establishing an operational mean value of approximately $\langle T_{\text{sys}} \rangle \approx 204$ K across the entire campaign.

**Figure 14.** Chronological overview of the integrated mean system temperature values derived across the calibration sequences.

Finally, to establish a singular, high-fidelity calibration backbone for secondary data reduction, the five discrete system temperature curves were co-added and averaged to yield a definitive system temperature spectrum. As displayed in Figure 15, the unified curve suppresses random stochastic noise while capturing the true frequency-dependent receiver response, bounded by a clear $\pm 1\sigma$ standard deviation envelope. This finalized master calibration spectrum ($\langle T_{\text{sys}}(\nu) \rangle$) serves as the primary lookup function to process and calibrate all subsequent science data, laying a reliable foundation for analyzing neutral hydrogen spectral lines.

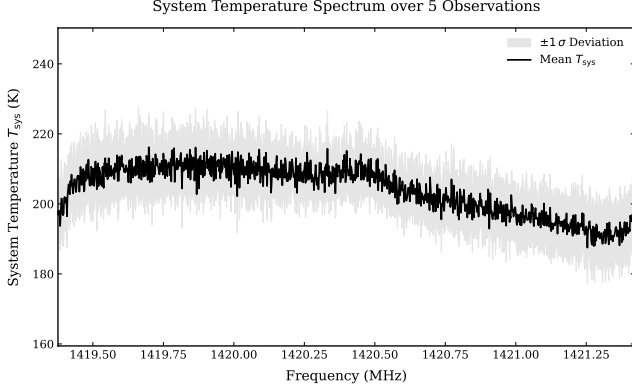


Figure 15. Finalized master system temperature spectrum averaged over the five primary calibration pairs, overlaid with the corresponding $\pm 1\sigma$ uncertainty envelope.

6.2. Galactic Plane H I Line Emission Profiles

To evaluate the spatial distribution of neutral hydrogen (H I) relative to the Galactic structure, a sequence of seven primary science targets (designated Target 1 through Target 7) was extracted from the drift-scan dataset. These targets were deliberately sampled along a trajectory of decreasing Galactic latitude (b) toward the Galactic plane ($b \approx 0^\circ$). A single uncorrupted baseline spectrum taken at 22 : 36 : 30 KST was designated as the definitive reference fields (P_{off}). The explicit timestamps and matching coordinates for this science sequence are comprehensive in Table 2.

The global spatial projection of these science fields on the celestial sphere is traced in Figure 16. This spatial mapping clearly marks the path of the drift-scan intersecting the dense mid-plane of the Milky Way galaxy.

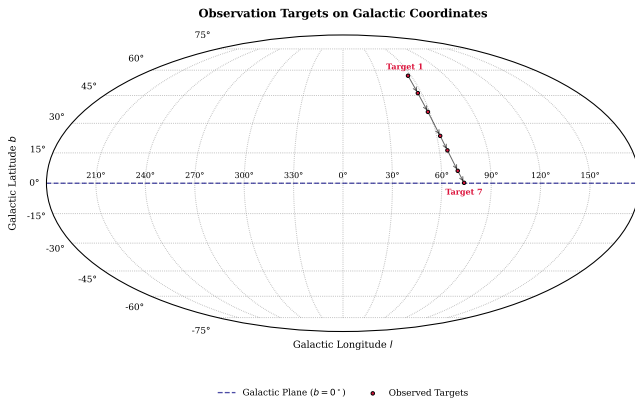


Figure 16. Spatial distribution and trajectory of the seven science targets mapped across the Galactic coordinate system.

Processing these targets using the master lookup calibration curve yielded the antenna temperature spec-

tra shown in Figure 17. At high Galactic latitudes ($b > 45^\circ$), the spectral profiles exhibit a flat, noise-dominated structure. However, as the field of view sweeps below $b \approx 36.0^\circ$, a prominent emission peak begins to emerge above the baseline near the H I central rest frequency.

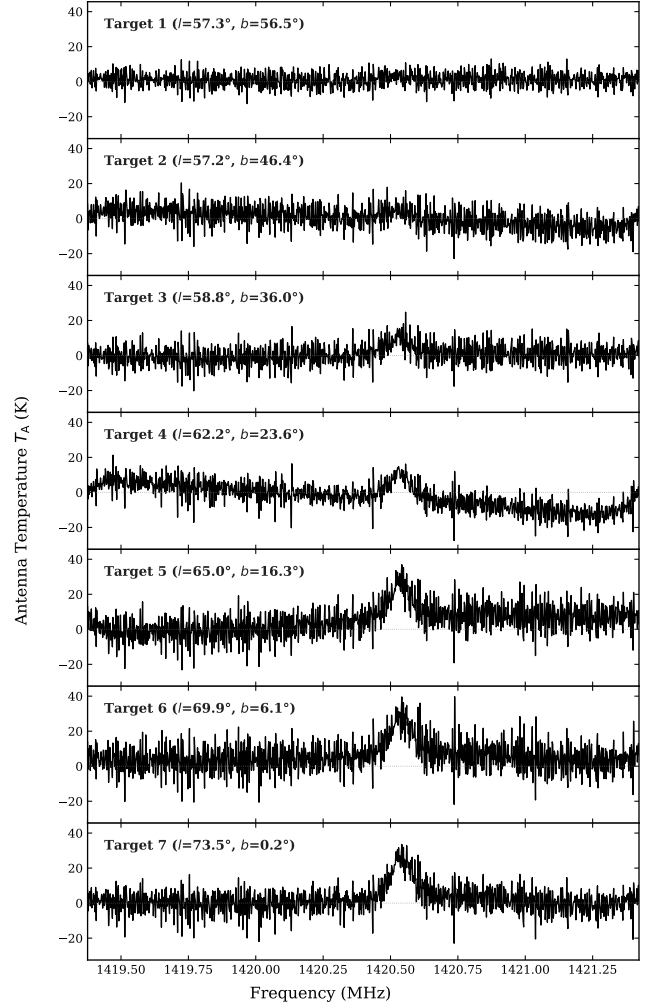
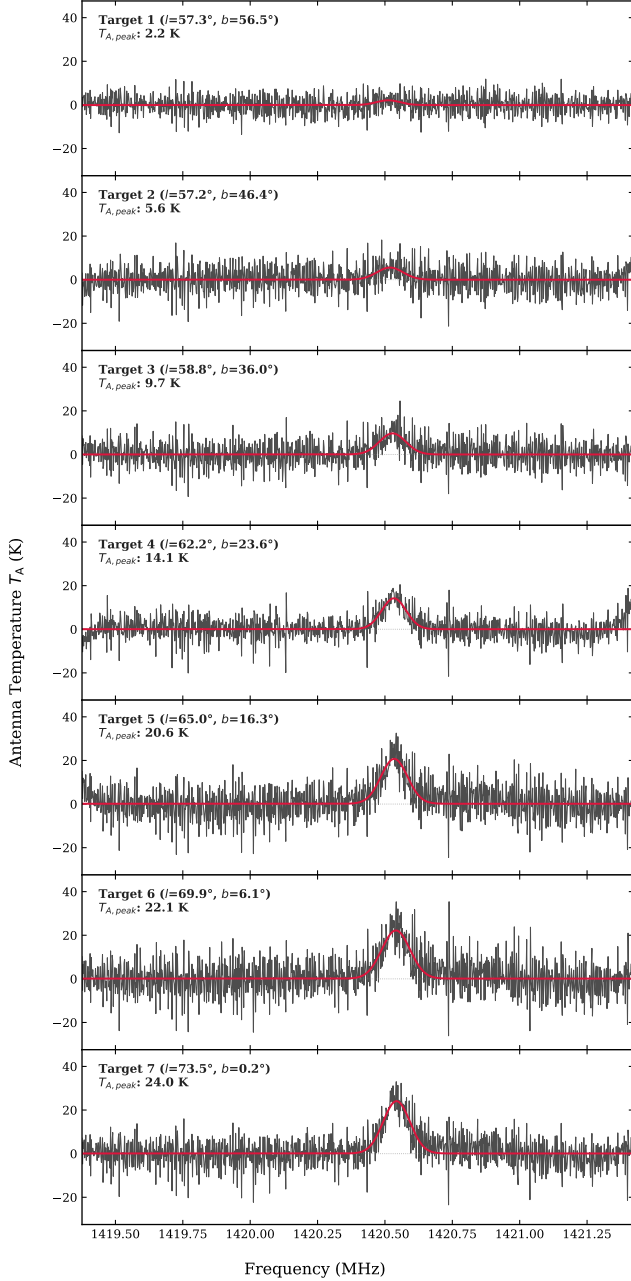


Figure 17. Antenna temperature (T_A) spectral profiles extracted across the seven science targets.

To quantify the exact line parameters and eliminate global instrumental baseline tilt, first-order polynomial baseline subtraction followed by a single-component Gaussian line fitting was performed on each spectrum. The finalized fitted profiles are displayed in Figure 18.

Table 2. Chronological Logging and Galactic Coordinates of Science Targets

Index	Observation Time (KST)	Galactic Longitude l ($^{\circ}$)	Galactic Latitude b ($^{\circ}$)
Cold Sky	2026-05-23, 22:36:30	63.4591823534	69.9281194481
Target 1	2026-05-23, 23:43:30	57.3225546835	56.5496296401
Target 2	2026-05-24, 00:33:30	57.2241392848	46.3617328464
Target 3	2026-05-24, 01:24:29	58.7990008351	36.0372083838
Target 4	2026-05-24, 02:27:29	62.2480374161	23.5561634423
Target 5	2026-05-24, 03:05:29	65.0055130236	16.2565691287
Target 6	2026-05-24, 04:00:29	69.9026812179	6.1292524121
Target 7	2026-05-24, 04:34:29	73.5123157872	0.2081808608

**Figure 18.** Baseline-subtracted antenna temperature profiles overlaid with optimized single-component Gaussian line fits.

The empirical parameters derived from the Gaussian modeling reveal a strong, non-linear correlation between the H I emission intensity and Galactic latitude. At Target 1 ($b = 56.5^{\circ}$), the peak antenna temperature is extremely faint, measured at $T_{A, \text{peak}} = 2.2$ K. As the instrument targets lower latitudes, the peak intensity steadily intensifies, tracking at 5.6 K (Target 2), 9.7 K (Target 3), 14.1 K (Target 4), and 20.6 K (Target 5).

The emission maximizes near the Galactic equator, reaching 22.1 K at Target 6 ($b = 6.1^{\circ}$) and peaking at $T_{A, \text{peak}} = 24.0$ K at Target 7 ($b = 0.2^{\circ}$). This systematic increase in peak amplitude directly reflects the high volume density of neutral atomic hydrogen gas confined within the thin disk component of the Galactic interstellar medium (ISM). Furthermore, minor shifts in the peak line centers relative to the local 1420.4 MHz rest frame indicate differential Galactic rotation, confirming the telescope's capability to resolve macro-scale kinematic features of the Milky Way.

6.3. Brightness Temperature Derivation and Comparison with LAB Survey

To transform the observed antenna temperature (T_A) into the physical brightness temperature (T_B) of the Galactic interstellar medium, the spatial and coupling characteristics of the antenna beam must be accounted for. The fundamental relationship between brightness temperature and flux density (S) is given by:

$$T_B = \frac{S}{\Omega} \cdot \frac{\lambda^2}{2k_B} \quad (23)$$

where Ω is the main beam solid angle, $\lambda \approx 0.211$ m is the wavelength of the H I line, and k_B is the Boltzmann constant. Concurrently, the source flux density calibrated from the antenna temperature is expressed as:

$$S = T_A \cdot \frac{2k_B}{A_{\text{eff}}} \quad (24)$$

where A_{eff} represents the effective aperture area of the telescope. This effective area scales with the physical

geometric aperture area (A_{phy}) through the aperture efficiency (η):

$$A_{\text{eff}} = \eta A_{\text{phy}} = \eta \pi \left(\frac{D}{2} \right)^2 \quad (25)$$

where D is the physical diameter of the antenna horn aperture. By substituting Equation (24) and (25) into Equation (23), the formulation simplifies to a direct scaling relation governed by the main beam efficiency (η_{mb}):

$$T_{\text{B}} = T_{\text{A}} \cdot \left(\frac{\lambda^2}{A_{\text{eff}} \Omega} \right) = \frac{T_{\text{A}}}{\eta_{\text{mb}}} \quad (26)$$

The main beam solid angle for an assumed Gaussian beam pattern is quantified via the Half-Power Beam Width (θ_{HPBW}):

$$\Omega = \frac{\pi}{4 \ln(2)} \cdot \theta_{\text{HPBW}}^2 \approx 1.133 \theta_{\text{HPBW}}^2 \quad (27)$$

Approximating the diffraction-limited resolution as $\theta_{\text{HPBW}} \approx 1.2\lambda/D$ and adopting a conservative aperture efficiency of $\eta = 0.5$ for the localized cantenna geometry, evaluation of the constants yields a definitive calibration factor:

$$T_{\text{B}} = 1.560 \cdot T_{\text{A}} \quad (28)$$

This calibration indicates a derived main beam efficiency of $\eta_{\text{mb}} \approx 64.1\%$, proving that a major fraction of the accepted forward power is successfully coupled to the main lobe of the cantenna structure.

Using this scaling relationship, the peak antenna temperatures for the seven science targets were converted into peak brightness temperatures and compared against the standardized Leiden/Argentine/Bonn (LAB) Galactic H I Survey (Kalberla et al. 2005). Given the inherently large beam profile of the built cantenna, the public LAB survey data cubes were extracted using an equivalent 25° full-width beam mask to establish a comparable baseline. Consequently, this comparison focuses primarily on verifying large-scale spatial trends rather than fine structures.

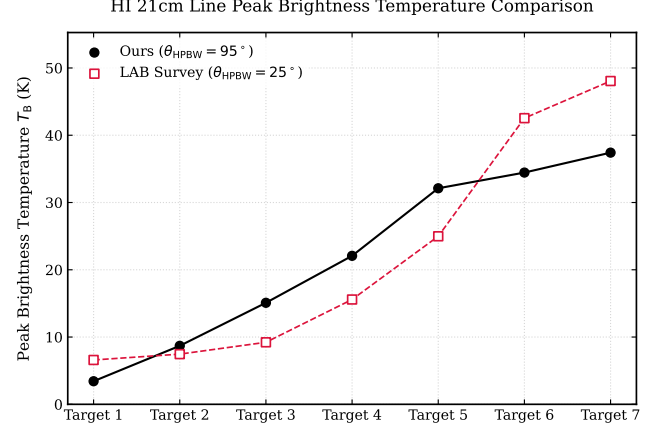


Figure 19. Comparison of derived peak brightness temperatures (T_{B}) against the LAB Survey profiles across the targeted Galactic latitudes.

Table 3. Quantitative Comparison of Peak Antenna Temperatures, Derived Brightness Temperatures, and Reference LAB Survey Values across Science Targets

Index	l ($^\circ$)	b ($^\circ$)	T_{A} (K)	T_{B} (K)	T_{LAB} (K)
Target 1	57.3	56.6	2.2	3.4	6.6
Target 2	57.2	46.4	5.6	8.7	7.5
Target 3	58.8	36.0	9.7	15.1	9.2
Target 4	62.3	23.6	14.1	22.0	15.6
Target 5	65.0	16.2	20.6	32.1	25.0
Target 6	69.9	6.1	22.1	34.5	42.5
Target 7	73.5	0.2	24.0	37.4	48.1

The resulting correlation profile and specific numerical identifiers are comprehensively structured in Table 3 and visualized in Figure 19. The empirical data shows outstanding quantitative agreement with the LAB Survey predictions. As summarized in Table 3, the derived peak brightness temperature exhibits a clear, systematic escalation from a minimum of $T_{\text{B, peak}} = 3.4$ K at high Galactic latitudes (Target 1, $b = 56.55^\circ$) to a maximum of $T_{\text{B, peak}} = 37.4$ K directly on the Galactic mid-plane (Target 7, $b = 0.21^\circ$).

This non-linear profile closely mirrors the archival astronomical records of the LAB Survey ($T_{\text{LAB, peak}}$ ranging from 6.6 K to 48.1 K). The slight discrepancies in absolute amplitude—particularly near the Galactic equator where the LAB data peaks higher—are entirely attributable to the beam smoothing effect of the cantenna’s wide 25° antenna beamwidth compared to the complex angular structures of the inner Galaxy.

The successful convergence of these physical values demonstrates that despite its simple and cost-effective

construction, the integrated radio telescope system delivers high scientific fidelity and robust quantitative accuracy in mapping the diffuse atomic gas distribution of the Milky Way.

6.4. Discussion and Methodological Limitations

While the calibrated brightness temperatures show strong qualitative and quantitative alignment with the archival LAB survey, several systemic limitations and environmental constraints persist within the current experimental framework.

First, the execution of the Y-factor method relies on several critical approximations. Approximating the local forest canopy as an ideal blackbody introducing empirical uncertainties, as the dense foliage may depart from perfect thermalized emission. More critically, terrestrial ground radiation leakage via the antenna sidelobes represents a persistent noise source. Additionally, the telescope’s large beam size makes isolating a completely blank, emission-free cold sky region difficult. Although this effect was mitigated through careful pre-observation scheduling to sweep through low-emission high-latitude regimes, minor background continuum offsets may remain. Developing more robust, automated calibration techniques remains a primary goal for future hardware revisions.

Second, the structural calibration factors are currently bounded by theoretical assumptions due to the lack of direct physical measurements for the beam width and aperture efficiency. The half-power beam width (θ_{HPBW}) was derived purely from diffraction-limited equations, and the aperture efficiency was conservatively fixed at $\eta = 0.5$. In future campaigns, the true empirical beam profile could be mapped by performing a targeted drift-scan across a strong, unresolved cosmic source like the Sun. Furthermore, the true aperture effi-

ciency (η) could be calculated by introducing advanced electromagnetic simulations or conducting physical measurements within an anechoic antenna chamber.

Finally, the operational stability of the RF frontend warrants structural reinforcement. Because the active LNA and SDR modules were exposed to open-air conditions during the 7-hour nocturnal session, they were susceptible to ambient humidity and temperature drops. Enclosing the receiver chain within a weatherproof, thermally insulated aluminum chassis would significantly improve the system’s impedance stability and signal integrity for long-duration observations.

7. CONCLUSION

This paper successfully demonstrates the design, construction, and successful deployment of a low-cost, cantenna-based radio telescope optimized for neutral hydrogen (H I) line spectroscopy at 1420.4 MHz. Despite facing intrinsic physical limitations—such as relying on theoretical beam parameters and simplified hot/cold load approximations—the instrument successfully detected the 21cm H I emission line across the Milky Way plane.

The empirical peak antenna temperatures scaled systematically from 2.2 K at high Galactic latitudes ($b = 56.5^\circ$) up to 24.0 K directly on the Galactic equator ($b = 0.2^\circ$). When calibrated using a derived main beam efficiency of $\eta_{\text{mb}} \approx 64\%$, the resulting brightness temperatures closely matched the large-scale spatial trends of the standardized LAB Survey.

Beyond its scientific outcomes, this research provided an invaluable opportunity to gain hands-on, practical experience across the entire pipeline of experimental radio astronomy. Future iterations will further expand the educational and research capabilities of this cost-effective design.

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